

two normalized same

multiple measures



unequal mass

different space

Extensions of OT

- So far: OT between **two normalized** measures on the **same** space.
- What happens when we relax these assumptions?
 - two → **multiple measures**
 - normalized → measures with **unequal mass**
 - same → **different space**
 - Also, extra constraints, new geometries, new data types.

Motivation: Why Extend OT?

Limitation	Extension	Examples
2 measures only	Multimarginal OT	Barycenters
Equal total mass	Unbalanced OT	Cell data, noisy histograms
Fixed coupling structure	OT + constraints	Finance (Martingale)
Same ambient space	Gromov–Wasserstein	Shape / graph matching
Scalar measures	Vector/matrix OT	Tensors, covariances

All preserve the variational flavor of OT!